Machine Learning vs. Deep Learning

Two powerful paradigms — same goal, different strengths. Explore how each approach is applied to solve real-world problems through two industry-defining use cases.

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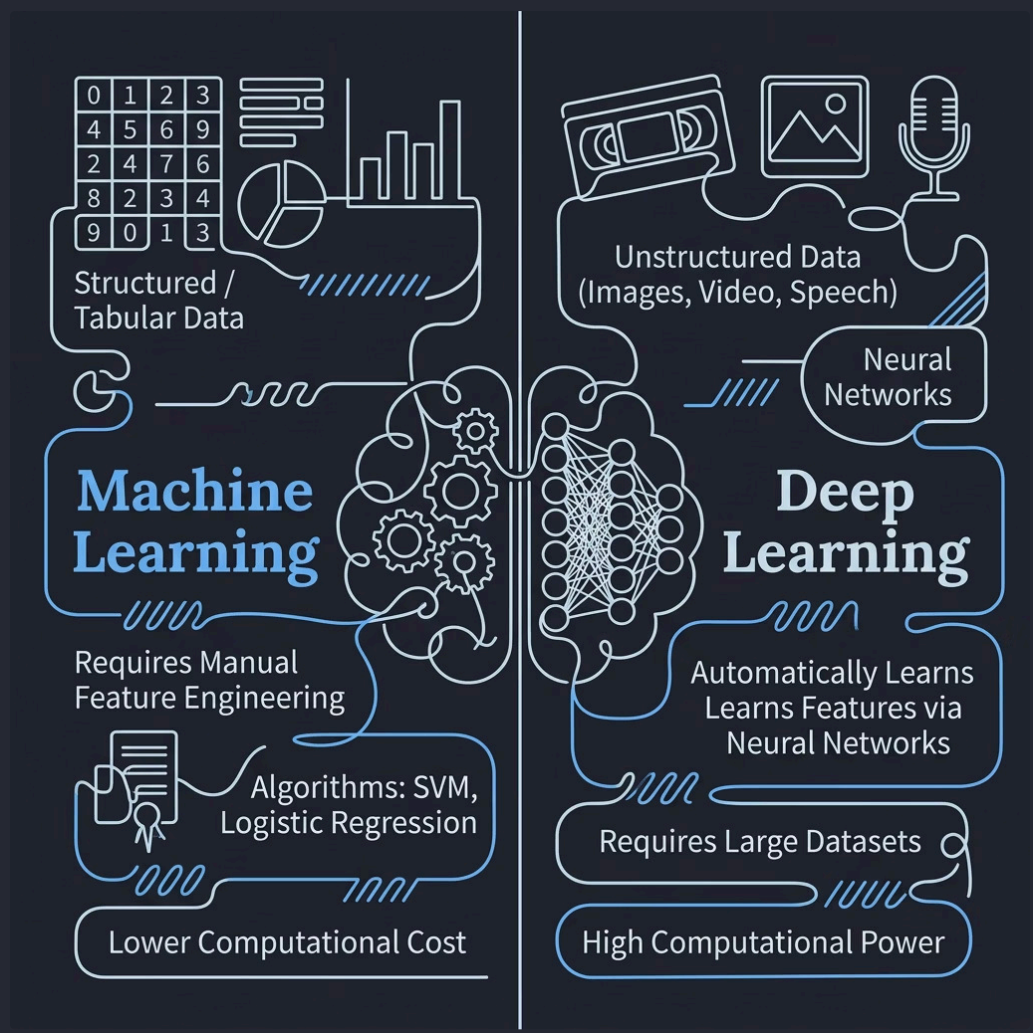
THE CORE DISTINCTION

What Sets Them Apart?



Both machine learning and deep learning are subfields of artificial intelligence — but they differ fundamentally in **how they learn** and **what data they handle best**.

Machine Learning

Excels with structured, tabular data. Features are engineered manually by domain experts.

Deep Learning

Excels with unstructured data. Neural networks learn hierarchical features automatically from raw input.

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USE CASE 01 — MACHINE LEARNING

Customer Churn Prediction at Spotify

Spotify uses ML models to identify subscribers at risk of canceling — and intervene before they leave.

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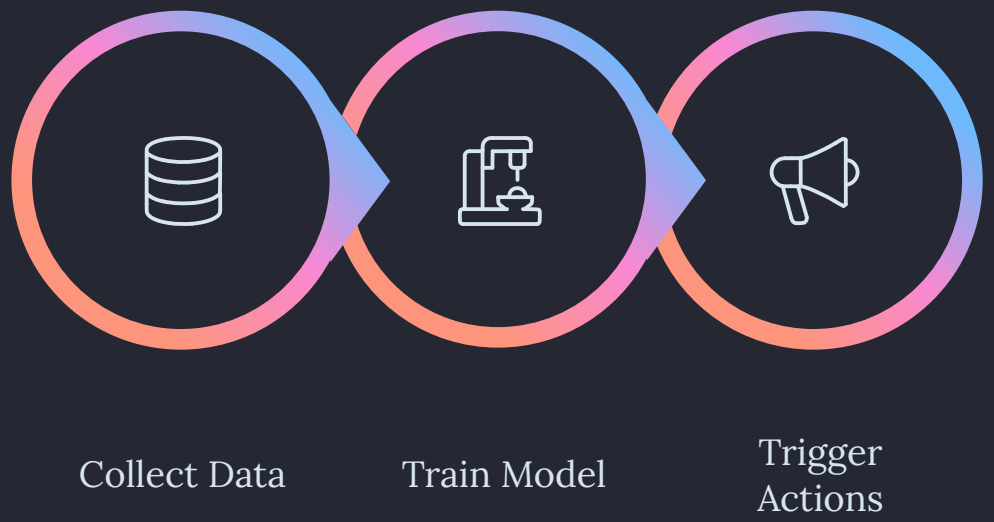
USE CASE 01 — MACHINE LEARNING

How Spotify Predicts Churn



Why Machine Learning Works Here

User behavior data — listening frequency, session duration, playlist activity — is inherently **structured and numerical**. This makes it a natural fit for classical ML algorithms.

Structured Input

Tabular engagement metrics convert cleanly into feature vectors.

Efficient Algorithms

SVM and Logistic Regression deliver strong predictive accuracy at low computational cost.

No Deep Learning Needed

The problem does not involve unstructured data — adding deep learning would add complexity without meaningful gain.

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USE CASE 02 — DEEP LEARNING

Autonomous Driving at Tesla

Tesla's self-driving system processes streams of complex visual and sensor data in real time — a task only deep learning can handle reliably.

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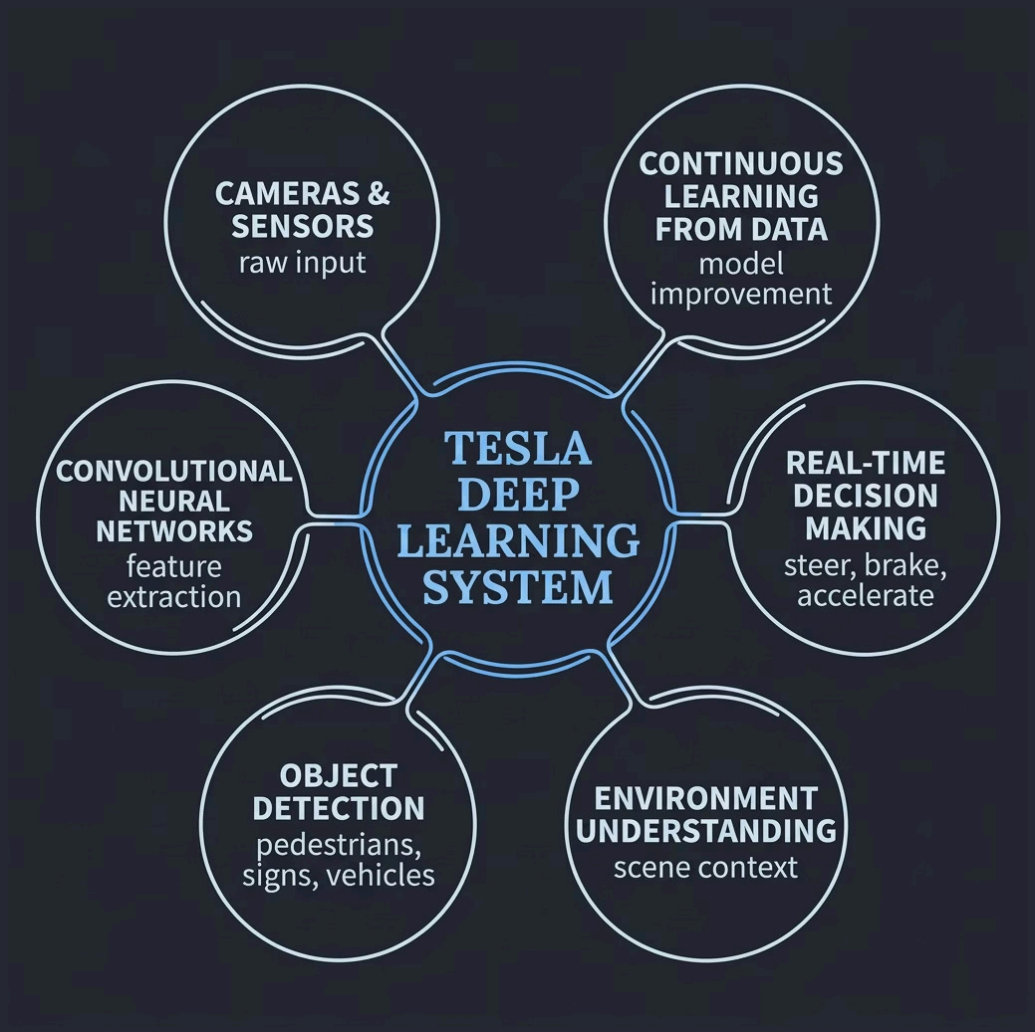
USE CASE 02 — DEEP LEARNING

How Tesla's System Sees the Road



Why Deep Learning Is Essential

Autonomous driving involves **high-dimensional, unstructured data** from cameras and sensors — exactly the domain where deep learning outperforms classical ML.

CNNs for Perception

Convolutional Neural Networks automatically detect objects — pedestrians, signs, vehicles — directly from raw image data.

Automatic Feature Learning

Unlike ML, no engineer needs to manually define shapes or edges. The model learns representations from millions of training frames.

Real-Time Decisions

Steering, braking, and lane adjustments are inferred instantly from a continuously changing environment.

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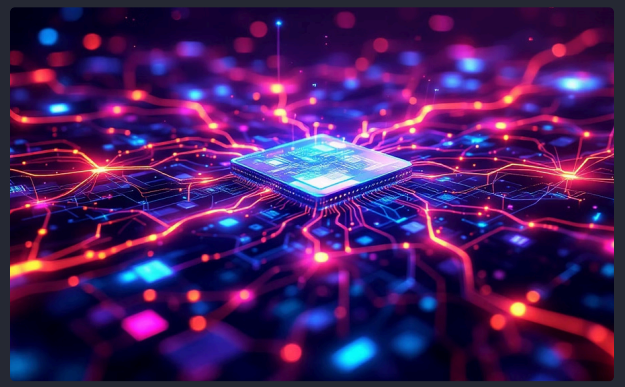
COMPARATIVE SUMMARY

Side-by-Side: When to Use Which



Machine Learning

- Structured, tabular input data
- Manual feature engineering
- Algorithms: SVM, Logistic Regression, Decision Trees
- Lower compute cost, faster training
- Best for: churn prediction, fraud detection, classification

Deep Learning

- Unstructured input: images, video, audio, text
- Automatic hierarchical feature learning
- Architectures: CNNs, RNNs, Transformers
- High compute cost, large dataset requirements
- Best for: computer vision, speech, autonomous systems

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Key Takeaways

1

Match the Tool to the Data

Structured data favors classical ML; unstructured data demands deep learning's representational power.

2

Complexity Has a Cost

Deep learning requires significantly more data and compute. Don't over-engineer — ML often suffices for well-defined, structured problems.

3

Real-World Validation

Spotify and Tesla illustrate how industry leaders select the right paradigm based on problem structure, not trend or preference.

Design principle: The best model is not always the most complex one — it is the one best suited to the nature of the problem.

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References

1. Spotify Engineering. (2021). How Spotify uses machine learning to create the future of personalization. https://engineering.atspotify.com/2021/12/how-spotify-uses-ml-to-create-the-future-of-personalization
2. Regami Solutions. (2025). Autonomous driving with AI-equipped vision systems: Tesla Autopilot. https://www.regami.solutions/post/autonomous-driving-ai-equipped-vision-systems-tesla-autopilot
3. Tesla. (2026). Full self-driving (supervised). https://www.tesla.com/support/fs

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